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Inspection Technology: Observable vs. Unobservable Choice

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Abstract In many contexts, a principal, such as a government, relies on inspectors or auditors to monitor companies or individuals to minimize the risks of adverse outcomes, such as environmental hazards or bank failures. In addition to effort, technology has become a crucial factor in this process. Advancements in technology, particularly artificial intelligence, have enhanced the productivity of monitoring efforts. However, rapid technological changes have also increased informational asymmetries regarding the technological capabilities being employed. For example, artificial intelligence washing (the practice of misrepresenting artificial intelligence capabilities to various stakeholders) has become increasingly common and has drawn regulatory scrutiny. This paper examines how the observability of technology influences the inspector's choice of technology in the inspector-inspectee strategic relationship, as well as the associated risk of adverse outcomes, which can be of primary concern to the principal. While prior literature explored cases where technology is unobservable but pre-determined, less is known about situations where both the inspector's choice of technology and effort are unobservable. The results suggest that while higher-capacity technology always reduces the probability of adverse outcomes when technology is observable, the same may not hold *ex-ante* when technology choice is unobservable. Additionally, when the choice of technology is unobservable, incentives (such as rewards and penalties set by the principal) may be less effective in influencing the inspector's technology choices.

Keywords Inspection · Technology · Choice · Imperfect · Information

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Introduction

A principal, such as a government, often seeks to minimize the probability of adverse outcomes, such as environmental hazards or bank failures, through inspector or auditor monitoring. This monitoring is influenced by both the inspector's effort and the technology used in the process. Dependence on technology, such as artificial intelligence (AI), has increased over time as it plays a greater role in enhancing productivity. However, rapid technological advancements have also increased informational asymmetry, making it more difficult to verify the technological capabilities employed. For example, it has become increasingly common for companies to exaggerate the extent of AI used in their products or services. This practice, known as AI washing (Prisznyák, 2023), has drawn regulatory attention. Recently, the United States (U.S.) Securities and Exchange Commission (SEC) charged two investment advisory companies with AI washing for falsely claiming to use advanced AI technologies (SEC, 2024).

As informational asymmetries regarding the inspector's choice of technology have increased, the strategic interaction between the inspector and the inspectee has become more complex, with important implications for the principal. This shift has transformed the problem from a one-sided asymmetric information issue, where only the inspectee's choices are unobservable and well-studied, into a two-sided asymmetric information problem, where both the inspector's capabilities and the inspectee's actions are unobservable (Kotowski et al., 2014). For example, while a bank regulator may not observe whether a bank is truthfully reporting the performance of its loan portfolio, the bank may also be unaware of the regulator's ability to detect discrepancies. Similarly, while an environmental compliance inspector may not directly observe a company's pollution levels, the company may be uncertain about the inspector's investigative capabilities. Kotowski et al. (2014) examined this two-sided asymmetric information problem in a setting where both the inspectee's actions and the inspector's predetermined technological capacity (high or low) were unobservable. This paper extends that inquiry by analyzing a two-sided asymmetric information problem in which, rather than having an unobservable but pre-determined technology level, the inspector's choice of inspection technology remains unobservable.

The paper reviews the relevant literature, highlighting key differences between previous studies and this work. Next, a strategic model of the inspector-inspectee relationship is introduced, in which the inspector selects both the level of inspection technology and effort, while the inspectee decides whether to shirk. The principal determines penalties and rewards for both players based on inspection findings or the occurrence of an adverse outcome, which may result from undetected shirking. The cases of observable and unobservable technology choices are then compared, evaluating their impact on the effectiveness of rewards and penalties as well as the likelihood of adverse outcomes.

Related Literature

This paper contributes to the literature on inspection games, which analyze the strategic interaction between an inspectee, who can choose to shirk, and an inspector, who determines the intensity of inspection that affects the likelihood of detecting shirking. These games have been applied in various areas, including auditing, crime, environmental monitoring, and tax inspections (Andreozzi, 2004; Avenhaus et al., 2002; Friehe, 2008).

Recent applications of inspection games incorporate real-life complexities and reflect policy or legal developments. For example, Dechenaux and Samuel (2019) compared announced versus unannounced inspections on compliance. Christmann and Klein (2024) examined how different liability regimes affect corporate crime, monitoring, and enforcement. Merlone et al. (2024) introduced a dynamic framework to analyze stability and optimal outcomes in organizational settings. Guzman et al. (2022) studied a sequential inspection game with limited visits using static and dynamic Stackelberg models. Boone and Dahan (2025) applied inspection games to security agencies, analyzing attacker-defender interactions for resource allocation and detection optimization. Zis (2021) used this framework to model global sulphur compliance and enforcement. Additionally, experimental studies, such as Rauhut and Winter (2021), explored how increased punishment severity influences norm enforcement and violations, challenging traditional deterrence theories.

In all these games, the inspection technology is either pre-determined or selected by the principal or the government. However, in many contexts, such as private auditing or situations where the principal's preferred technology cannot be enforced, the inspector may independently choose the technology level.[1] For example, AI use in auditing is rising (Fedyk et al., 2022), where different AI types (ranging from task automation to context-aware intelligence (Hasan, 2021)) entail varied costs, such as the need for costly IT specialists with advanced technology. These differences, combined with unobservable levels of technological intelligence and human effort, create the two-sided asymmetric information problem mentioned earlier.

The distinction between a pre-determined, unobservable technology type and unobservable choices of technology type is important. Kotowski et al. (2014) showed that penalties and rewards can drive separation between high- and low-technology types when these types are fixed. However, this paper demonstrates that the same incentives may be less effective when technology choice is unobservable. Specifically, under observable technology, if the inspector lacks incentives to choose high-capacity technology, the principal can influence this choice by adjusting rewards and penalties. In the unobservable case, for a range of parameter values, technology choice remains unaffected by the inspector's rewards and penalties, making these less effective. Under observable technology, the adverse outcome (important for the principal) is least likely under high-capacity technology and can be avoided through adjusting the inspector's incentives. In contrast, when technology

[1] The government may provide advanced technology, but the inspector might still opt for older technology if it remains available, unobservable, and cheaper to use.

choice is unobservable, higher-capacity technology does not always yield the lowest probability of an adverse outcome. Notably, the study finds that under unobservable technology, any technology used in equilibrium by the inspector may minimize the probability of the adverse outcome, which, to the best of our knowledge, is a novel result.

The Model

At date 1, the inspector (*IR*) chooses between high (*H*) and low (*L*) productivity inspection technologies. At date 2, the inspectee (*IE*) plays shirk (*S*) or does not play shirk (*DS*) and *IR* plays inspect (*I*) or does not play inspect (*DI*). The technology choice can be observable by all or by *IR* only. The players' date 2 actions are unobservable to each other.

S can lead to an adverse outcome with probability $(1-q)$. For simplicity, it is assumed that the adverse outcome cannot occur under *DS*. The adverse outcome is costly for both players: C_A^{IE} and C_A^{IR} for *IE* and *IR*, respectively.[2] B_S^{IE} is *IE*'s expected benefit under *S* when the adverse outcome does not occur. It is assumed that under *DI*, *S* is preferred to *NS*, i.e.

$$\pi_S^{IE} = qB_S^{IE} - (1-q)C_A^{IE} > \pi_{DS}^{IE} = 0.$$

Inspection is imperfect. *S* is undetected if *DI* is played. p and bp are the detection probabilities under *H* and *L* respectively, $0 < b < 1$. *H* has a non-recoverable fixed part F_H and a variable part V_H. F_H is incurred when *H* is adopted, even if not used. V_H is incurred only when *H* is used. The corresponding costs for *L* are F_L and V_L. If *S* is detected, penalty C_P^{IE} is imposed on *IE*.[3] The reward to *IR* is B^{IR}. For example, C_P^{IE} can be a fine or the cost of reversing *S*. B^{IR} could be a bonus or another benefit for detecting and preventing *S*, which could have led to an adverse outcome.[4] Figure 1 summarizes the model..

Inspector's Technology Choice is Observable

First, the case of observable choice is considered. Table 1 shows the resulting date 2 subgames. The following assumptions ensure that there is no pure strategy NE, as is the case in standard inspection games.

[2] The adverse outcome can hurt *IR*'s reputation, as seen in the Enron scandal. Enron's risky bets were missed by Arthur Andersen, its auditor. When the adverse outcome occurred, Enron collapsed. Arthur Andersen went out of business due to its damaged reputation.

[3] It is assumed that *IR* cannot frame *IE* when there is no shirking.

[4] For simplicity, it is assumed that *IR* gets a reward (benefit) only if *S* is detected. If the adverse outcome does not occur for other reasons, *IR*'s benefit is zero. Allowing a benefit in these cases will not change the results.

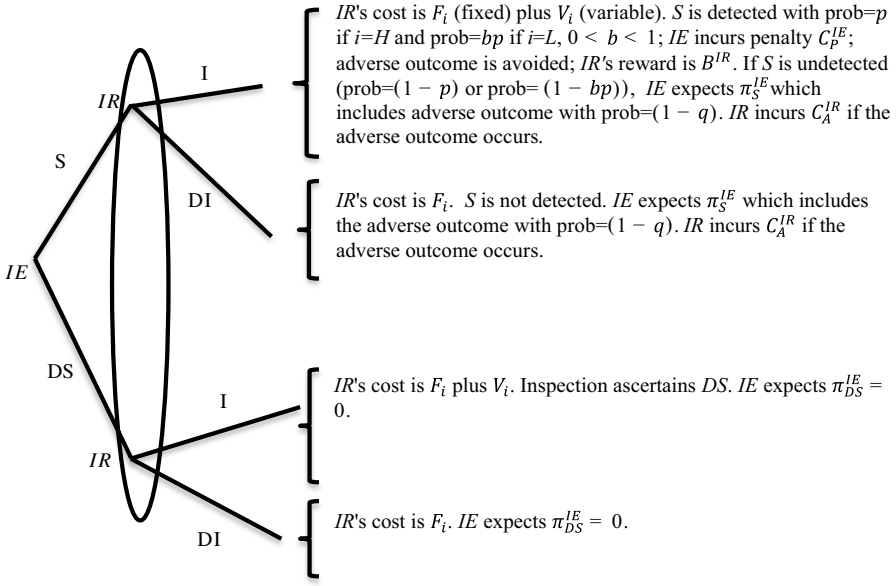


Fig. 1 Model Summary. $i = H, L$ productivity technology is chosen at date 1

Table 1 Observable technology choice. Date 2 subgames. H subgame if $i = H$; L subgame if $i = L$

		Inspector (*IR*)	
		I	*DI*
Inspectee (*IE*)	*S*	$-pC_P^{IE} + (1-p)\pi_S^{IE}$ if i=H $-bpC_P^{IE} + (1-bp)\pi_S^{IE}$ if i=L $-F_H - V_H + pB^{IR}$ $-(1-p)(1-q)C_A^{IR}$ if i=H $-F_L - V_L + bpB^{IR}$ $-(1-bp)(1-q)C_A^{IR}$ if i=L	π_S^{IE} $-(1-q)C_A^{IR} - F_H$ if i=H $-(1-q)C_A^{IR} - F_L$ if i=L
	DS	0 $-F_H - V_H$ if i=H $-F_L - V_L$ if i=L	0 $-F_H$ if i=H $-F_L$ if i=L

Entries in the upper-left part of each cell show the row player's payoffs, while those in the lowerright part show the column player's payoffs

IE's best response to *I* is *DS*:[5]

$$-pC_p^{IE} + (1-p)\pi_S^{IE} < 0, \tag{1}$$

[5] If $-bpC_P^{IE} + (1-bp)\pi_S^{IE} > 0$, S is a strictly dominant strategy under L but not under H, which is not true in typical inspection games. For this reason and for the sake of brevity, results for this case are not reported.

$$-bpC_p^{IE} + (1-bp)\pi_S^{IE} < 0. \tag{2}$$

IR's best response to *S* is *I*:

$$\begin{array}{c} -V_H + pB^{IR} - (1-p)(1-q)C_A^{IR} > -(1-q)C_A^{IR}, \\ -V_L + bpB^{IR} - (1-bp)(1-q)C_A^{IR} > -(1-q)C_A^{IR} \end{array} \tag{3}$$

Proposition 1 describes the subgame perfect equilibrium of the observable choice.

Proposition 1. *(i) If* $F_H - F_L < \frac{(1-q)C_A^{IR}}{p(B^{IR}+(1-q)C_A^{IR})}\left(\frac{V_L}{b} - V_H\right)$, *IR will choose H:*

$$prob(S) = \frac{V_H}{p(B^{IR} + (1-q)C_A^{IR})},$$

$$prob(I) = \frac{qB_S^{IE} - (1-q)C_A^{IE}}{p(C_P^{IE} + \pi_S^{IE})}.$$

(ii) Otherwise, IR will choose L:

$$prob(S) = \frac{V_L}{bp(B^{IR} + (1-q)C_A^{IR})},$$

$$prob(I) = \frac{qB_S^{IE} - (1-q)C_A^{IE}}{bp(C_P^{IE} + \pi_S^{IE})}.$$

Proof. In the mixed equilibrium, when *H* is played,

$$\begin{array}{c} prob(S)\left(-F_H - V_H + pB^{IR} - (1-p)(1-q)C_A^{IR}\right) + (1-prob(S))\left(-F_H - V_H\right) = \\ prob(S)\left(-F_H - (1-q)C_A^{IR}\right) + (1-prob(S))\left(-F_H\right), \\ prob(I)\left(-pC_P^{IE} + (1-p)\pi_S^{IE}\right) + (1-prob(I))\left(\pi_S^{IE}\right) = 0. \end{array}$$

The solutions yield $prob(S)$ and $prob(I)$.

IR'*s* expected payoff is

$$E_H^{IR} = -F_H - \frac{V_H(1-q)C_A^{IR}}{p(B^{IR} + (1-q)C_A^{IR})}. \tag{4}$$

In the mixed equilibrium, when *L* is played,

$$\begin{array}{c} prob(S)\left(-F_L - V_L + \mathrm{b}pB^{IR} - (1-\ \mathrm{b}p)(1-q)C_A^{IR}\right) + (1-prob(S))\left(-F_L - V_L\right) \\ = prob(S)\left(-F_L - (1-q)C_A^{IR}\right) + (1-prob(S))\left(-F_L\right), \\ prob(I)\left(-\mathrm{b}pC_P^{IE} + (1-\mathrm{b}p)\pi_S^{IE}\right) + (1-prob(I))\left(\pi_S^{IE}\right) = 0. \end{array}$$

The solutions yield $prob(S)$ and $prob(I)$.

IR'*s* expected payoff is

$$E_L^{IR} = -F_L - \frac{V_L(1-q)C_A^{IR}}{pb(B^{IR} + (1-q)C_A^{IR})}. \tag{5}$$

H is chosen if Eq. (4) > Eq. (5) or

$$F_H - F_L < \frac{(1-q)C_A^{IR}}{p\left(B^{IR} + (1-q)C_A^{IR}\right)}\left(\frac{V_L}{b} - V_H\right). \tag{6}$$

If Eq. (6) holds with equality, *IR* breaks the tie by choosing *L*. End of Proof.

According to Eq. (6), *H is* not chosen if $\frac{V_L}{b} - V_H < 0$ while $F_H - F_L > 0$, *t*he latter being a reasonable assumption. To allow choice, it is assumed that

$$\frac{V_L}{b} - V_H > 0. \tag{7}$$

Given Eq. (7), it can be checked that *prob(I)* and *prob(S)* are smaller under *H*. According to Eq. (6), technology choice will depend on cost and productivity differences.

*T*he choice will also depend on C_A^{IR} and B^{IR}, *i*n contrast to the unobservable case in the next section. It can be verified that the righthand side of Eq. (6) will increase in C_A^{IR}, increasing incentives to switch to *H*. However, decreasing rather than increasing B^{IR} can increase *IR*'*s* incentive to choose *H*.

Inspector's Technology Choice is Unobservable

Now *IR*'s actual choice is unobservable. Since $F_H > F_L$, (*L*, *DI*) will strictly dominate (*H*, *DI*) and the game will be reduced to a 2 × 3 game as shown in Table 2.

Equation (3) from the observable case is modified to

$$-F_H - V_H + pB^{IR} - (1-p)(1-q)C_A^{IR} > -(1-q)C_A^{IR} - F_L, \tag{8}$$

In addition, if $F_H + V_H < F_L + V_L$, (*H, I*) would strictly dominate (*L, I*), resulting in a simple 2 × 2 game with a mixture of (*H,I*) with (*L,DI*) only. To keep the discussion interesting, it is assumed that

$$-F_H - V_H < -F_L - V_L. \tag{9}$$

According to Eq. (8) and Eq. (9), *IR* will prefer (*L,I*) over (*H,I*) under *DS* and (*H,I*) over (*L,I*) under *S*. The unique mixed equilibrium of this game is described in Proposition 2.

Table 2 Unobservable technology choice

Inspector (*IR*)

Inspectee (*IE*)	*H, I*	*L, I*	*L, DI*
S	$-pC_P^{IE} + (1-p)\pi_S^{IE}$ $-F_H - V_H + pB^{IR} - (1-p)(1-q)C_A^{IR}$	$-bpC_P^{IE} + (1-bp)\pi_S^{IE}$ $-F_L - V_L + bpB^{IR} - (1-bp)(1-q)C_A^{IR}$	π_S^{IE} $-(1-q)C_A^{IR} - F_L$
DS	0 $-F_H - V_H$	0 $-F_L - V_L$	0 $-F_L$

Entries in the upper-left part of each cell show the row player's payoffs, while those in the lower-right part show the column player's payoffs

Proposition 2. *(i) If* $F_H - F_L < \left(\frac{V_L}{b} - V_H\right)$*, the unique, mixed strategy Nash Equilibrium is*

$$prob(H, I) = \frac{qB_S^{IE} - (1-q)C_A^{IE}}{p\left(C_P^{IE} + \pi_S^{IE}\right)},$$

$$prob(H, DI) = 0,$$

$$prob(L, DI) = 1 - prob(H, I),$$

$$prob(L, I) = 0,$$

$$prob(S) = \frac{F_H + V_H - F_L}{p\left(B^{IR} + (1-q)C_A^{IR}\right)}.$$

(ii) If $F_H - F_L > \left(\frac{V_L}{b} - V_H\right)$*, the unique, mixed strategy Nash Equilibrium is*

$$prob(H, I) = 0,$$

$$prob(H, DI) = 0,$$

$$prob(L, I) = \frac{qB_S^{IE} - (1-q)C_A^{IE}}{bp(C_P^{IE} + \pi_S^{IE})},$$

$$prob(L, DI) = 1 - prob(L, I),$$

$$prob(S) = \frac{V_L}{bp(B^{IR} + (1-q)C_A^{IR})}.$$

Proof. Given Eq. (1) and Eq. (2), *DS* is a best response to (*H,I*) and (*L,I*), so *IR*'*s* (*L,DI*) must be played with a non-zero probability in any mixed equilibrium. Then, any pure strategy played by *IR* in a mixed equilibrium should have the same expected payoff as (*L,DI*).

Comparing the expected payoffs from (*H,I*) and (*L,I*),

$$\begin{aligned} &prob(S)\left(-F_H - V_H + pB^{IR} - (1-p)(1-q)C_A^{IR}\right) \\ &+ (1 - prob(S))\left(-F_H - V_H\right) \geq prob(S)\left(-F_L - V_L + \mathrm{b}pB^{IR} - (1 - \mathrm{b}p)(1-q)C_A^{IR}\right) \\ &+ (1 - prob(S))\left(-F_L - V_L\right), \end{aligned}$$

or $prob(S) \geq \frac{F_H+V_H-F_L-V_L}{p(B^{IR}+(1-q)C_A^{IR})}$.

The last inequality implies a threshold *prob(S)* above which *(H, I)* is mixed with *(L, DI)*. Below this threshold, *(L,I)* is mixed with *(L, DI)*. A continuum of equilibria exists at equality.

The mixed equilibrium with *(L,I)* and *(L,DI)* is the same as the one for the *L* subgame of Proposition 1. *IR*'*s* expected payoff is

$$-F_L - \frac{V_L}{bp(B^{IR} + (1-q)C_A^{IR})}(1-q)C_A^{IR}. \tag{10}$$

The mixed equilibrium with *(H, I)* and *(L,DI)* is found from Table 2 by eliminating *(L,I)*. The solution yields $prob(S)$, $prob(H, I)$ and $prob(L, DI)$. *IR*'*s* expected payoff is

$$-F_L - \frac{F_H + V_H - F_L}{p(B^{IR} + (1-q)C_A^{IR})}(1-q)C_A^{IR}. \tag{11}$$

The mixture with H is chosen if Eq. (11) > Eq. (10) oxr $F_H - F_L < \left(\frac{V_L}{b} - V_H\right)$. End of Proof.

If $F_H - F_L = \left(\frac{V_L}{b} - V_H\right)$, $prob(S) = \frac{F_H+V_H-F_L}{p(B^{IR}+(1-q)C_A^{IR})} = \frac{V_L}{bp(B^{IR}+(1-q)C_A^{IR})}$ and *IR*'s equilibrium mixture is indeterminate. The expected payoff lines from *IR*'s three pure strategies against *IE*'s choice of *prob(S)* will intersect at the same point. Since this can only arise for exceptional combinations of payoffs, it is relatively unimportant and not considered.

According to Proposition 2, if the cost difference between *H* and *L* is large enough, irrespective of the players'rewards and penalties, *H* will not be used. Recall that in the observable case, for the same cost difference, *H* can be chosen if *IR*'s penalty is large enough.

In the second equilibrium of Proposition 2, *L* is played with certainty with resulting *Prob(S)* and *Prob(I)* identical to Proposition 1. In the first equilibrium of Proposition 2, both technologies have a positive probability of being chosen.

To see why *IR*'s rewards and penalties do not matter now, consider initial parameter values such that *L* is played with certainty in both observable and unobservable cases. The equilibrium probabilities of *S*, *I* and expected payoffs would be the same in both cases. Then C_A^{IR} increases such that Eq. (6) becomes true and *IR* switches to *H* in the observable case due to a higher expected payoff. In the unobservable case, switching to *H* will also increase *IR*'s payoff if *H* were credible. However, because technology is unobservable, the commitment to *H* is non-credible. If *IE* believes that *H* is chosen and *IR* is inspecting with the probability of Proposition 1, *IE* would shirk with the probability of Proposition 1. However, then *IR* would play (*L*, *DI*), to which *IE*'s best response is *S*. As long as $F_H - F_L > \left(\frac{V_L}{b} - V_H\right)$ and the previously made assumptions hold, a change in C_A^{IR} will not disrupt the equilibrium in Proposition 2. Similarly, a change in B^{IR} will not affect the technology choice.

Additionally, once the cost difference is small enough so that H is played with a positive probability, this probability can be increased through the other player's rewards and penalties, as suggested by $prob(H,I) = \frac{qB_S^{IE} - (1-q)C_A^{IE}}{p\left(C_P^{IE} + \pi_S^{IE}\right)}$. This probability will increase if B_S^{IE} increases or if C_A^{IE} or C_P^{IE} decrease. These are standard results from inspection games, where a player's equilibrium probability depends on the opponent's payoffs.

Probability of the Adverse Outcome

The probability of the adverse outcome is

$$P_A = prob(S)(1 - prob(I)prob(DetectS/I))(1-q).$$

(i) Observable technology
Under H,

$$P_A = \frac{V_H}{p\left(B^{IR} + (1-q)C_A^{IR}\right)}\left(1 - \frac{\pi_S^{IE}}{\left(C_P^{IE} + \pi_S^{IE}\right)}\right)(1-q). \tag{12}$$

Under L,

$$P_A = \frac{V_L}{bp\left(B^{IR} + (1-q)C_A^{IR}\right)}\left(1 - \frac{\pi_S^{IE}}{\left(C_P^{IE} + \pi_S^{IE}\right)}\right)(1-q). \tag{13}$$

Equation (12) Eq. (13) if $\frac{V_L}{b} - V_H > 0$. In the observable case, P_A is lower under H, making it the principal's preferred choice. H can be induced by increasing IR's penalty or decreasing his reward, as discussed earlier.

(ii) Unobservable Technology
In the mixed equilibrium with H and L,

$$P_A = \frac{F_H + V_H - F_L}{p\left(B^{IR} + (1-q)C_A^{IR}\right)}\left(1 - \frac{\pi_S^{IE}}{\left(C_P^{IE} + \pi_S^{IE}\right)}\right)(1-q). \tag{14}$$

In the equilibrium with L only, P_A is the same as in Eq. (13). Equation (14) $<$ Eq. (13) if $F_H - F_L < \left(\frac{V_L}{b} - V_H\right)$, which corresponds to the first equilibrium of Proposition 2. When the second equilibrium is played, Eq. (14) $>$ Eq. (13). Thus, whichever equilibrium is played in the unobservable case, it also yields the lowest P_A.

Conclusion

Our results suggest that incentives affecting technology adoption in inspection games can depend on the observability of the technology. With symmetric observability, the inspector's adoption incentives can be directly affected through his expected rewards and penalties. When the inspector alone observes the technology, his rewards and penalties may be less effective for technology adoption.

The study finds that in the case of symmetric observability, the adverse outcome will have a lower probability under the more advanced technology, making it the principal's preferred choice. In the case of asymmetric observability, *ex ante*, more advanced technology does not necessarily lead to the lowest probability of adverse outcome. Whichever equilibrium is played in this case, it also yields the lowest probability of the adverse outcome.

A possible limitation of the paper is the one-shot nature of the game. Extending the model to a repeated interaction with an uncertain end period would be a natural next step.

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References

Andreozzi, L. (2004). Rewarding policemen increases crime. Another surprising result from the inspection game. *Public Choice, 121*(1), 69–82.

Avenhaus, R., Von Stengel, B., & Zamir, S. (2002). Inspection games. *Handbook of game theory with economic applications, 3*(Chapter 51), 1947–1987.

Boone, J. H. & Dahan, M. (2025). Inspection game with imperfect detection technology. *Naval Research Logistics (NRL)*. https://doi.org/10.1002/nav.22241

Christmann, R., & Klein, D. (2024). Game theory, compliance, and corporate criminal liability: Insights from a three-player inspection game. *Decision Analytics Journal, 11*(June), 100431.

Dechenaux, E., & Samuel, A. (2019). Announced or surprise inspections and oligopoly competition. *The BE Journal of Economic Analysis & Policy, 19*(1), 1–20.

Fedyk, A., Hodson, J., Khimich, N., & Fedyk, T. (2022). Is artificial intelligence improving the audit process? *Review of Accounting Studies, 27*(3), 938–985.

Friehe, T. (2008). Correlated payoffs in the inspection game: Some theory and an application to corruption. *Public Choice, 137*(1), 127–143.

Guzman, C., Riffo, J., Telha, C., & Van Vyve, M. (2022). A sequential Stackelberg game for dynamic inspection problems. *European Journal of Operational Research, 302*(2), 727–739.

Hasan, A. R. (2021). Artificial intelligence (AI) in accounting & auditing: A literature review. *Open Journal of Business and Management, 10*(1), 440–465.

Kotowski, M. H., Weisbach, D. A., & Zeckhauser, R. J. (2014). Audits as signals. *University of Chicago Law Review, Essay, 81*(1), 179.

Merlone, U., Orlando, G., & Dal Forno, A. (2024). Dynamical inspection game with continuous strategies. *Annals of Operations Research, 337*(3), 1205–1234.

Prisznyák, A. (2023). Horizontal and vertical value creation in bankrobotics and the AI-washing phenomenon. *Financial and Economic Review, 22*(3), 97–122.

Rauhut, H., Winter, F. (2021). How More Severe Punishment Generates Less Norm Enforcement. Further Evidence of Paradoxical Effects of Norm Enforcement in Inspection Games. In Krumpal, I., Raub, W., Tutić, A. (eds) Rationality in Social Science, pp. 149–173. Springer, Wiesbaden. https://doi.org/10.1007/978-3-658-33536-6_8

U.S. Securities and Exchange Commission (SEC). (2024). SEC charges two investment advisers with making false and misleading statements about their use of artificial intelligence. Available from: https://www.sec.gov/newsroom/press-releases/2024-36. Accessed 8 Mar 2025.

Zis, T. P. (2021). A game theoretic approach on improving sulphur compliance. *Transport Policy, 114*, (December) 127–137.

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